Advanced Programming Practice · 21CSC203P

Unit 2 · Session 9 · SLO 1 — Solved Worksheet

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Unit – 2 : Session – 9 : SLO – 1 | Polymorphism Classification

■ Activity – Mystery Boxes (Answers)

A. area() in both Circle and Rectangle → RUN-TIME POLYMORPHISM (if they share a base class/interface and override area(); resolved by object type at run time)

B. Printer with print(String) and print(int) → COMPILE-TIME POLYMORPHISM (method overloading — chosen by argument types at compile time)

C. Loop printing "Hello" five times → NO POLYMORPHISM (plain iteration)

D. Subclass overrides speak() from superclass → RUN-TIME POLYMORPHISM (overriding)

• E. Method that adds two numbers → NO POLYMORPHISM (ordinary method)

• F. Animal reference pointing to a Dog object calling makeSound() → RUN-TIME POLYMORPHISM (dynamic dispatch through base reference)